

(34)
CO2- AT3
Slot 'D'

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Code of Conduct

I Abdul Razak Shaikh (192524353) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abdul Razak Shaikh

cloud Auto Scaling and load Balancing

A. A company may experience sudden traffic spikes during seasonal sales such as Black Friday or festive offers. cloud auto-scaling automatically adds more servers when traffic increases and removes them when demand decreases. load balancing distributes incoming user requests across multiple servers, preventing any single server from becoming overloaded.

This approach reduces downtime because if one server fails or becomes busy, the load balancer redirects traffic to healthy servers. As a result, users experience faster page loading, fewer errors, and uninterrupted shopping.

Monitoring tools continuously track CPU usage, memory, response time, and traffic. They trigger auto scaling when predefined thresholds are reached.

Examples- During an Amazon Prime Daysale, millions of customers visit the website simultaneously. Auto Scaling launches additional servers, while Load Balancers distribute traffic evenly, ensuring smooth shopping without service interruption.

2. Cloud Storage for Backup Data.

cloud storage services provide high durability by storing multiple copies of data across different disks and data centers. This redundancy ensures that data remains available even if hardware fails.

Sensitive backup data is protected using encryption and access control through user authentication, permissions and multi-factor authentication (MFA). These security measures prevent unauthorized access.

Compared to traditional storage systems, cloud storage is more cost-effective because organizations pay only for the storage they use, eliminating the need to purchase and maintain expensive hardware.

Example:- If a company's data center is damaged by a flood, backups stored in the cloud can quickly restore systems, minimizing downtime and data loss.

Global Application Deployment Using CDNs.

A Content Delivery Network (CDN) stores copies of application content on servers located around the world. Users receive data from the nearest server, reducing latency and improving performance.

Edge servers process requests close to users resulting in faster loading times, lower delays, and better user experience across different regions.

Cloud providers offer global availability through multiple regions and availability zones, ensuring high reliability and fast content delivery worldwide.

Examples- A video streaming platform like Netflix or an online gaming service delivers content from nearby edge servers, reducing buffering and gameplay lag for users in different countries.

4. Cloud security and Compliance for financial Institutions

Cloud providers support regulatory compliance by offering certifications such as ISO 27001, PCI DSS, GDPR and SOC standards. They also provide auditing features to track user activities and system changes.

Logging records system events, monitoring detects suspicious activities and encryption protects financial data during storage and transmission. Together, these features strengthen security.

To meet data sovereignty requirements, organizations can store customer data in specific geographic regions, as required by local laws.

Example:- A bank processing online payments stores customer financial records in an approved regional data center while encrypting all transactions and maintaining detailed audit logs for regulatory inspection.

2 Containerization and Kubernetes

Containers package an application and its dependencies into a lightweight unit that shares the host operating system. Unlike VMs, Containers do not require a separate operating system for each instance, making them faster to start and more efficient in resource usage.

Kubernetes simplifies deployment by automatically managing containers, balancing workloads, scaling applications, restarting failed containers and performing rolling updates.

Examples- An e-commerce application built with microservices may have separate containers for the user interface, product catalog, payment service and order management. Kubernetes manage these containers ensuring reliable deployment and automatic scaling as user demand changes.